AutoVet AI Forecast: Benchmark Guide

What to write in Chapter 4 (Results and Discussion) and Chapter 5 (Conclusions and Recommendations), explained in simple terms with examples.
Data window: 24 October 2025 to 24 October 2026 · Prepared 25 October 2026

Before you start: the AI in one page

What the AI does

For every inventory item, AutoVet looks at how many units were used each day. It adds those up into a **running total** (total units used so far). Then it draws the **best straight line** through those points. This is **linear regression**, done with the scikit-learn library.

The **slope** of that line is the most important number. It tells us **how many units the clinic uses per day**. With that, AutoVet predicts when the item will run low:

```
Days until stock-out = (current stock − minimum stock) ÷ units used per day
```

> **Example.** Pet Shampoo has 39 bottles in stock. The minimum is 10, so 29 bottles can be used before it is "low". The AI learned that the clinic uses about 0.23 bottles a day. So 29 ÷ 0.23 ≈ **126 days** until it reaches the minimum.

Words you will need

Word	Simple meaning
Slope (daily rate)	How many units are used per day, learned from the line.
80/20 holdout test	The AI learns from the oldest 80% of days and is tested on the newest 20% that it never saw. It is like studying past exams, then taking a new exam.
Error	Actual value minus predicted value. Smaller is better.
MAE (Mean Absolute Error)	The average size of the errors. "On average, the prediction was off by this many units."
RMSE (Root Mean Square Error)	Like MAE, but big misses count more. If RMSE is much bigger than MAE, a few predictions were far off.
R^2 (R-squared)	A score of how well the line fits. 1 = perfect; 0 = no better than guessing the average; below 0 = worse than guessing the average.
Benchmark	A measuring stick. We compare the AI with simple methods on the **same data** to see if the AI is worth using.
Baseline method	A simple, "no AI" way of forecasting that we compare against.
Median	The middle value when all items are lined up from smallest to largest. We use it instead of the average so that one strange item does not distort the result.

CHAPTER 4 · Results and Discussion

How the AI was tested, what the numbers mean, and how it compares with simple methods

4.1 How the AI was tested

Each item's usage history is split by date. The **oldest 80%** of days are for learning and the **newest 20%** are for testing. The AI never sees the test days while learning. It predicts them, and we compare its predictions with what really happened. An item needs at least 10 days with usage to be tested this way. Items with fewer days are only checked on the data they learned from.

This follows the "training set / test set" approach described in forecasting textbooks (Hyndman & Athanasopoulos, 2021; James et al., 2021).

4.2 Worked example: how one item is tested

To show every step, here is a small made-up example of a syringe used over 10 days. The numbers were run through AutoVet's own code, so they match what the system calculates.

Day	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
Used that day	2	3	2	3	3	4	3	5	4	5
Running total	2	5	7	10	13	17	20	25	29	34
Used for	Learning (80% = 8 days)								Testing (20%)	

1. **Learn.** The AI draws the best line through days 0–7. Its slope is **3.20 units per day**. The line fits the learning days very well (R^2 = 0.99, "High" confidence).
2. **Predict.** The last known total is 25 (day 7). The AI adds 3.20 per day: day 8 = 25 + 3.20 = **28.20**, and day 9 = 25 + 6.40 = **31.40**.
3. **Compare.** The real totals were 29 and 34. The errors are 0.80 and 2.60.
4. **Score.** MAE = (0.80 + 2.60) ÷ 2 = **1.70 units**. RMSE = $\sqrt{((0.80^2 + 2.60^2) \div 2)}$ = **1.92 units**. RMSE is a bit higher than MAE because the day-9 miss was bigger.

4.3 The simple methods we compare against (the benchmark)

A benchmark does **not** mean finding another system with the same AI. It means testing the AI and some simple methods on the **same items and the same test days** and comparing their errors. Forecasting textbooks recommend this: a method is only worth using if it beats the simple ones (Hyndman & Athanasopoulos, 2021). We used three simple methods. Each one guesses a daily usage from the learning days only:

Method	**How it guesses daily usage**	**In the syringe example**
Average method	Total used ÷ number of days so far	25 ÷ 8 = 3.13 per day
Naïve method	Same as the last day	Day 7 used 5, so 5 per day
7-day moving average	Average of the last 7 days	23 ÷ 7 = 3.29 per day
AutoVet AI	**Slope of the best-fit line**	**3.20 per day**

Like the AI, each method starts from the last known total (25) and adds its daily guess:

Method	Day 8 guess (real: 29)	Day 9 guess (real: 34)	MAE	RMSE
AutoVet AI	**28.20**	**31.40**	**1.70**	**1.92**
Average method	28.13	31.25	1.81	2.04
Naïve method	30.00	35.00	1.00	1.00
7-day moving average	28.29	31.57	1.57	1.79

> **What this example teaches.** In this example the AI beat the average method but not the naïve method, because usage happened to jump at the end. **No method wins on every item.** That is why the real benchmark tests every item and then looks at the overall picture.

4.4 How the test is scored: starting from the last known total

When AutoVet predicts a stock-out, it starts from **today's real stock** and uses only the daily rate (slope). So the test must score the AI the same way: start from the last known total and add the daily rate. That is what steps 2–4 above did.

An earlier version of the test started from the line's own starting point (its *intercept*). The system never uses that point to predict anything, so the test was grading a prediction the system does not make. In the syringe example, that version would guess 26.79 and 29.99 instead of 28.20 and 31.40, and its MAE would be 3.11 instead of 1.70. The test was corrected so that it measures what the system actually does.

> **Be transparent about this.** If the panel asks, explain that the formula did not change. Only the way the test is scored changed, so that it matches how the system predicts. Before the correction, the median MAE across items was 5.39. After it, the median MAE is 2.50.

4.5 Results on the AutoVet data

The data. There are 41 inventory items. 37 had at least 3 days of usage and could be forecast; 4 had too little data. 24 items had enough history (10 or more days) for the full 80/20 test. The data covers 24 October 2025 to 24 October 2026. Most usage records are **demo (synthetic) data** generated to resemble a real clinic, plus records from actual use of the system.

Table A. Benchmark on all data (24 items with an 80/20 test)

Method	Median MAE (units)	Median RMSE (units)	Items where the AI was more accurate (by MAE)
AutoVet AI (linear regression)	**2.50**	**2.78**	—
Average method	2.21	2.62	6 of 24
7-day moving average	3.83	4.49	14 of 24
Naïve method	12.18	14.09	20 of 24

Other results: the median fit on the learning days was R^2 = 0.98, and every forecast item was rated "High" confidence. The median test R^2 was **0.48**, which means the AI did better than simply guessing the average

of the test days.

Table B. Benchmark on real records only (demo data removed)

Method	Median MAE	Median RMSE	Items where the AI was more accurate (by MAE)
AutoVet AI (linear regression)	**5.89**	**6.41**	—
Average method	5.04	5.45	2 of 5
7-day moving average	11.30	13.27	2 of 5
Naïve method	42.38	54.44	4 of 5

With real records only, 22 items could be forecast but only 5 had enough history for the 80/20 test. Five items is too few to draw strong conclusions, so Table A is the main result and Table B is supporting.

4.6 A real item as an example

Pet Shampoo (ANA-944E58) has 35 days of usage; the newest 7 were used for testing. The AI's average error on those days was 0.57 bottles. The average method's was 0.64, the 7-day moving average's 1.88, and the naïve method's 10.86. Here the AI was the most accurate of all four. Its learned rate of about 0.23 bottles a day gives the 126-day stock-out estimate shown on page 1.

4.7 What the results mean

- **The AI clearly beats the naïve method** (20 of 24 items). Copying yesterday's usage is unreliable because one busy or quiet day throws the guess off. The AI learns from all the days, so one unusual day matters less.
- **The AI beats the 7-day moving average on most items** (14 of 24), because it uses more history than just the last week.
- **The average method is slightly more accurate overall** (2.21 vs 2.50). This is expected when usage is **steady**: if a clinic uses about the same amount every day, the long-run average is already a very good guess. Most of the data is demo data generated at a steady rate, which favours the average method.
- **Where the AI helps:** the AI and the average method give almost the same answer when usage is steady. When usage is rising or falling, the AI's line follows the change, while a plain average is held back by old days.

SAMPLE PARAGRAPH YOU CAN ADAPT FOR CHAPTER 4

The forecasting model was evaluated using an 80/20 holdout test, in which the oldest 80% of each item's usage records were used for training and the newest 20% for testing (Hyndman & Athanasopoulos, 2021; James et al., 2021). Test predictions were projected from the last known usage total, the same way the system computes days until stock-out. Accuracy was measured with MAE and RMSE; both were reported because MAE shows the typical error while RMSE gives more weight to large misses (Chai & Draxler, 2014). The model was benchmarked against three simple forecasting methods on the same 24 items and test days. It was more accurate than the naïve method on 20 of 24 items and than the 7-day moving average on 14 of 24 items. The average method achieved a slightly lower median error (MAE 2.21 vs 2.50 units). This is consistent with the mostly steady usage in the dataset, a condition under which the long-run average is a strong predictor. The model's median test R^2 of 0.48 shows that it predicted unseen days better than a simple mean.

CHAPTER 5 · Conclusions and Recommendations

What the results prove, what they do not prove, and what to do next

5.1 Summary of findings

- AutoVet's AI learns each item's daily usage from its history and predicts how many days remain before stock reaches the minimum level.
- On 24 items tested with an 80/20 holdout, the AI's median error was 2.50 units (MAE) and 2.78 units (RMSE), with a median test R^2 of 0.48.
- The AI was more accurate than the naïve method on 20 of 24 items and than the 7-day moving average on 14 of 24 items.
- The simple average method was slightly more accurate overall (median MAE 2.21), mainly because usage in the dataset was steady.
- Correcting the test so that it scores the AI the way the system actually predicts lowered the median error from 5.39 to 2.50 units.

5.2 Conclusions

In simple terms: **the AI gives useful and reasonably accurate stock-out warnings, and it is clearly better than simple guessing based on recent days.** It performs about as well as a long-run average on steady items. It does not beat every simple method on every item, and this study does not claim that it does.

SAMPLE PARAGRAPH YOU CAN ADAPT FOR CHAPTER 5

The study concludes that the linear regression forecasting model integrated into AutoVet provides reasonably accurate estimates of inventory depletion. In benchmark testing, the model outperformed the naïve and 7-day moving average methods on most items, and its accuracy was comparable to that of the simple average method. The average method's slight advantage is attributed to the steady usage patterns in the dataset. The results also show the importance of evaluating a model the way it is actually used: scoring predictions from the last known usage total gave a truer measure of the system's performance.

5.3 Recommendations

Recommendation	Why, in simple terms
1. Collect more real usage data	Only 5 items had enough real history for the full test. More months of real clinic use would make the results stronger and less dependent on demo data.
2. Re-run the benchmark regularly	The system already includes a benchmark command (`php artisan inventory:forecast-benchmark`). Running it every few months shows whether the AI stays accurate as real data grows.
3. Try methods that handle seasons and trends	Some items may be used more in certain months, such as vaccines during drives. Future researchers can try methods made for seasonal patterns, such as exponential smoothing (Hyndman & Athanasopoulos, 2021), and compare them with the same benchmark.
4. Consider combining the AI with the average method	The average was strongest on steady items and the AI follows changes better. A future version could use the average when usage is steady and the AI's trend when it is changing.
5. Add an error measure that compares items fairly	An error of 2 units is big for a slow item but small for a fast one. Scaled measures such as MASE make items comparable (Hyndman & Koehler, 2006).
6. Review the confidence and alert levels	The "High / Medium / Low" confidence cut-offs (R^2 of 0.70 and 0.40) and the 7- and 14-day alert levels were set by the researchers. They can be adjusted with feedback from clinic staff.

Possible panel questions and simple answers

Question	Simple answer
Why didn't your AI beat every method?	No method wins on every item. On steady usage the plain average is already a very good guess. Our AI matches it closely and clearly beats the other two simple methods. We report the results honestly instead of hiding the comparison.
Did you change the formula to get better numbers?	No. The prediction formula is the same. We corrected how the test is scored so that it starts from the last known total, which is exactly how the system predicts stock-outs.
Why use the median instead of the average?	A few items have unusual numbers. The median shows the typical item without being pulled by one extreme case.
Why report both MAE and RMSE?	MAE shows the typical error and RMSE shows whether there were big misses. Reporting both gives a fuller picture (Chai & Draxler, 2014).
Is the data real?	It is a mix of records from actual use of the system and demo data generated to resemble a clinic. We also ran the benchmark on real records only (Table B) and report it separately.
Why linear regression and not a more complex AI?	It is simple, fast, easy to explain to clinic staff, and works with the small amount of data a clinic has. More complex models need much more data.

References (APA 7th edition)

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Before citing, open each source and check the page or section you refer to. All numbers in this guide come from running AutoVet's benchmark command on the live database on 25 October 2026. If the data changes, re-run the command and update the tables.